Shamsiya Solar Cooling Umbrella System

Technical Specification Sheet

Model: SH-001
Date: February 10, 2026

Panel 1 — System Overview & Process Conditions

A) General Product Information			
Product Name	–	Shamsiya 3 × 3	
Model Name	–	SH-001	
System Type	–	Refrigeration-based air processing	
Regeneration Capability	–	Included (regeneration mode available)	
B) Physical Envelope			
Overall Dimensions (L×W×H)	m	3 × 3 × 3	
Total Covered Volume	m^3	27	
C) Electrical System			
Primary Energy Source	–	Solar PV	
Energy Storage Type	–	Lithium Iron Phosphate ($LiFePO_4$)	
Nominal Battery Voltage	V	48	
Battery Capacity	Ah	600	
Nominal Stored Energy	kWh	28.8	
Usable Energy (80% DoD)	kWh	23.0	
Battery Configuration	–	16S $LiFePO_4$ (48V class)	
Integrated Protection	–	BMS (cell balancing, OVP, UVP, OCP)	
Load Supply Mode	–	Simultaneous continuous operation	
D) Electrical / Utility Streams			
Compressor Power (Energy Stream)	kW	0.5556	(2000 kJ/h)
Electric Heater (Energy Stream)	kW	2.222	(7998 kJ/h)
Air Curtains (Electrical Load)	kW	2.000	(7200 kJ/h)
Fans (Electrical Load)	kW	1.000	(3600 kJ/h)
Total Electrical Demand	**kW**	**5.778**	
Total Energy Rate Equivalent	kJ/h	2.080×10^4	
E) Cold Air Inlet			
Phase	–	Vapour	
Temperature	°C	40.00	
Pressure	kPa	101.3	
Molar Flow Rate	kmol/h	2.810	
Mass Flow Rate	kg/h	80.00	
Std Volumetric Flow Rate	m^3/h	66.37	

Panel 2 — Architecture & Performance Summary

F) System Architecture (Flowsheet Components)		
Compression Unit	–	Refrigerant compressor (K-100)
Heat Exchange Stages	–	E-100, E-101, E-102, E-103
Expansion Device	–	VLV-100 (throttle valve)
Moisture Separation	–	V-100 (separator/receiver)
Drying / Filtration	–	X-100 (silica/desiccant, regenerable)
G) Hot Air Inlet		
Phase	–	Vapour
Temperature	°C	40.00
Pressure	kPa	101.3
Molar Flow Rate	kmol/h	10.54
Mass Flow Rate	kg/h	300.0
Std Volumetric Flow Rate	m^3/h	248.9
H) Target Air Outlet (Delivered)		
Phase	–	Vapour
Delivered Air Temperature	°C	21.33
Delivered Air Pressure	kPa	101.3
Molar Flow Rate	kmol/h	2.712
Mass Flow Rate	kg/h	78.23
Std Volumetric Flow Rate	m^3/h	64.07
I) Thermodynamic Indicators		
Cold Inlet Molar Enthalpy	kJ/kmol	-1.014×10^4
Cold Inlet Molar Entropy	kJ/(kmo	123.5
Cold Inlet Heat Flow	kJ/h	-2.850×10^4
Target Molar Enthalpy	kJ/kmol	-2326
Target Molar Entropy	kJ/(kmo	118.8
Target Heat Flow	kJ/h	-6306
J) Key Performance Summary		
Delivered Cooling (Temperature)	–	**40.00 °C → 21.33 °C**
Cooling Capacity (Air-Side, ΔH)	kW	**6.17**
Total Electrical Demand (All Loads)	kW	**5.778**
System COP (Cooling / Electrical)	–	**1.07**
Moisture Removal Rate	kg/h	**1.77**
K) Humidity State (101.3 kPa)		
Initial Condition	–	40.00 °C @ **60% RH**
Delivered Condition	–	21.33 °C @ **36% RH**

Basis: 101.3 kPa; $y_{H_2O,in} = 0.0437$, $y_{H_2O,out} = 0.0091$. Cooling from stream heat flows: $|-28500 - (-6306)| / 3600 = 6.17$ kW.

Notes: Prototype / simulation-based specifications. Electrical ratings and compliance certifications are design targets and may change with final hardware selection.

Contact: Shamsiya Engineering Team